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# UNIDOCK – STUDENT RESOURCE HUB

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ICT 1209 – Web Technologies  
Mini Project Final Report



**Theme: Student Resource Hub**

**GROUP MEMBERS:**

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DATE -09/26/2026

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# 1. Project Overview

## 1.1 Purpose

UniDock is a web based Student Resource Hub. The platform helps university students share, discover, and access academic resources such as lecture notes, past papers, and study guides, while also giving them simple tools to organize their day to day academic life.

## 1.2 Target Audience

The primary users are undergraduate students they have the facility to upload and access study material, track assignments and their deadlines, inquire about academic matters, and communicate with platform administrators.

## 1.3 Key Features

- Study Resource Upload – students can upload lecture notes, past papers and study guides as PDF files.
- Smart Search – keyword and subject based search across uploaded resources.
- Bookmarks – students can bookmark resources for quick access later.
- Interactive To Do List – add, complete, and delete daily academic tasks.
- Reminders – create and manage reminders for exams and assignment deadlines
- Ask a Question (Q&A) – students can post questions and reply to each others questions
- Contact Form – visitors and students can send messages to the platform team
- Responsive Design – built with Bootstrap 5 so the site adapts to mobile, tablet and desktop

# 2. System Design

## 2.1 Database Schema

### Users

| Field      | Type         | Key         | Description                          |
|------------|--------------|-------------|--------------------------------------|
| Id         | INT          | Primary Key | Unique user identifier               |
| Full name  | VARCHAR(100) |             | Student's full name                  |
| Email      | VARCHAR(150) | UNIQUE      | Login email address                  |
| Password   | VARCHAR(255) |             | Password hashed with password hash() |
| Created at | TIMESTAMP    |             | Account creation date                |

## Resources

| Field        | Type         | Key                    | Description                 |
|--------------|--------------|------------------------|-----------------------------|
| Id           | INT          | Primary Key            | Unique resource identifier  |
| Title        | VARCHAR(200) |                        | Title of the resource       |
| Subject      | VARCHAR(100) |                        | Subject / faculty           |
| Subject code | VARCHAR(50)  |                        | Optional subject code       |
| description  | VARCHAR(255) |                        | Short description           |
| File path    | VARCHAR(255) |                        | Path to the stored PDF file |
| Uploaded_by  | INT          | Foreign Key - users.id | Uploader (nullable)         |
| Created at   | TIMESTAMP    |                        | Upload date                 |

## Bookmarks

| Field       | Type      | Key                       | Description                         |
|-------------|-----------|---------------------------|-------------------------------------|
| Id          | INT       | Primary Key               | Unique bookmark identifier          |
| User_id     | INT       | Foreign Key - users.id    | Student who bookmarked the resource |
| Resource_id | INT       | Foreign Key -resources.id | Bookmarked resource                 |
| Created_at  | TIMESTAMP |                           | Bookmark date                       |

## Todos

| Field      | Type         | Key                   | Description            |
|------------|--------------|-----------------------|------------------------|
| Id         | INT          | Primary Key           | Unique todo identifier |
| User_id    | INT          | Foreign Key- users.id | Owner of the task      |
| Title      | VARCHAR(200) |                       | Task title             |
| Note       | TEXT         |                       | Optional note          |
| Created at | TIMESTAMP    |                       | Creation date          |

## Reminders

| Field      | Type         | Key                   | Description                |
|------------|--------------|-----------------------|----------------------------|
| Id         | INT          | Primary Key           | Unique reminder identifier |
| User_id    | INT          | Foreign Key- users.id | Owner of the reminder      |
| Title      | VARCHAR(200) |                       | Reminder title             |
| Note       | TEXT         |                       | Optional note              |
| Created_at | TIMESTAMP    |                       | Creation date              |

## Questions

| Field      | Type         | Key                   | Description                           |
|------------|--------------|-----------------------|---------------------------------------|
| Id         | INT          | Primary Key           | Unique question identifier            |
| User_id    | INT          | Foreign Key- users.id | Student who asked                     |
| Title      | VARCHAR(200) |                       | Auto-generated from the question body |
| Body       | TEXT         |                       | Full question text                    |
| Created_at | TIMESTAMP    |                       | Post date                             |

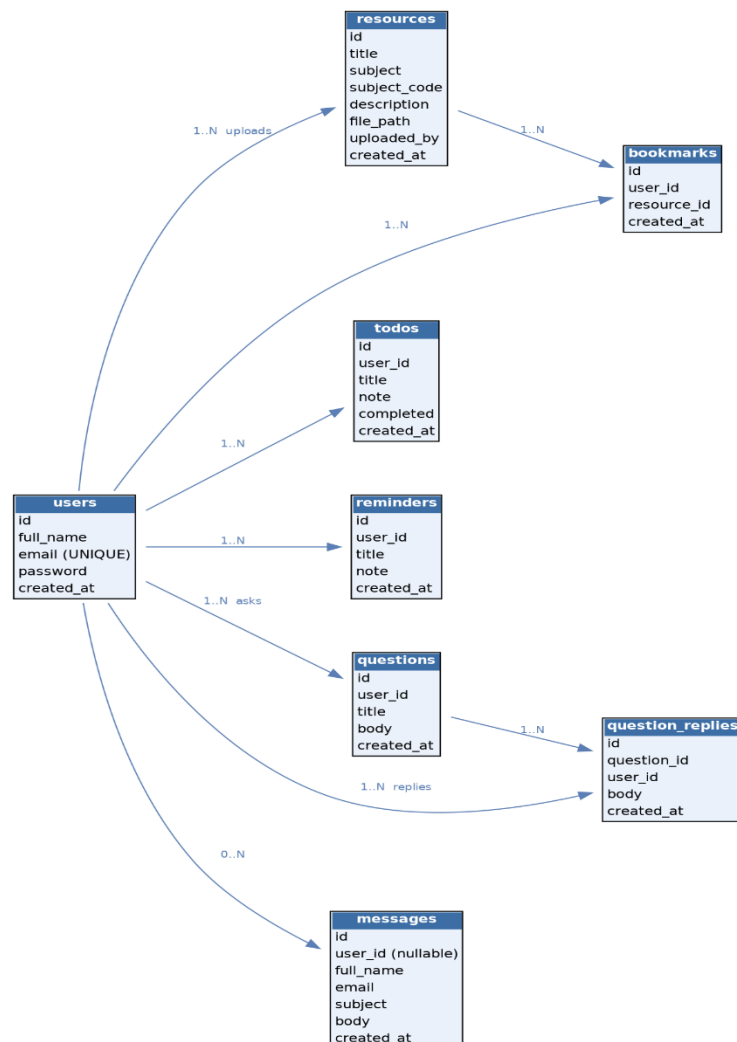
## Question replies

| Field       | Type      | Key                       | Description             |
|-------------|-----------|---------------------------|-------------------------|
| Id          | INT       | Primary Key               | Unique reply identifier |
| Question_id | INT       | Foreign Key- questions.id | Question being answered |
| User_id     | INT       | Foreign Key- users.id     | Student who replied     |
| Body        | TEXT      |                           | Reply text              |
| Created_at  | TIMESTAMP |                           | Reply date              |

## Messages

| Field      | Type         | Key                    | Description               |
|------------|--------------|------------------------|---------------------------|
| Id         | INT          | Primary Key            | Unique message identifier |
| User_id    | INT          | Foreign Key- users.id, | Sender, if logged in      |
| Full_name  | VARCHAR(100) |                        | Sender's name             |
| Email      | VARCHAR(150) |                        | Sender's email            |
| Subject    | VARCHAR(200) |                        | Optional subject          |
| Body       | TEXT         |                        | Message content           |
| Created_at | TIMESTAMP    |                        | Date sent                 |

## 2.2ER diagram



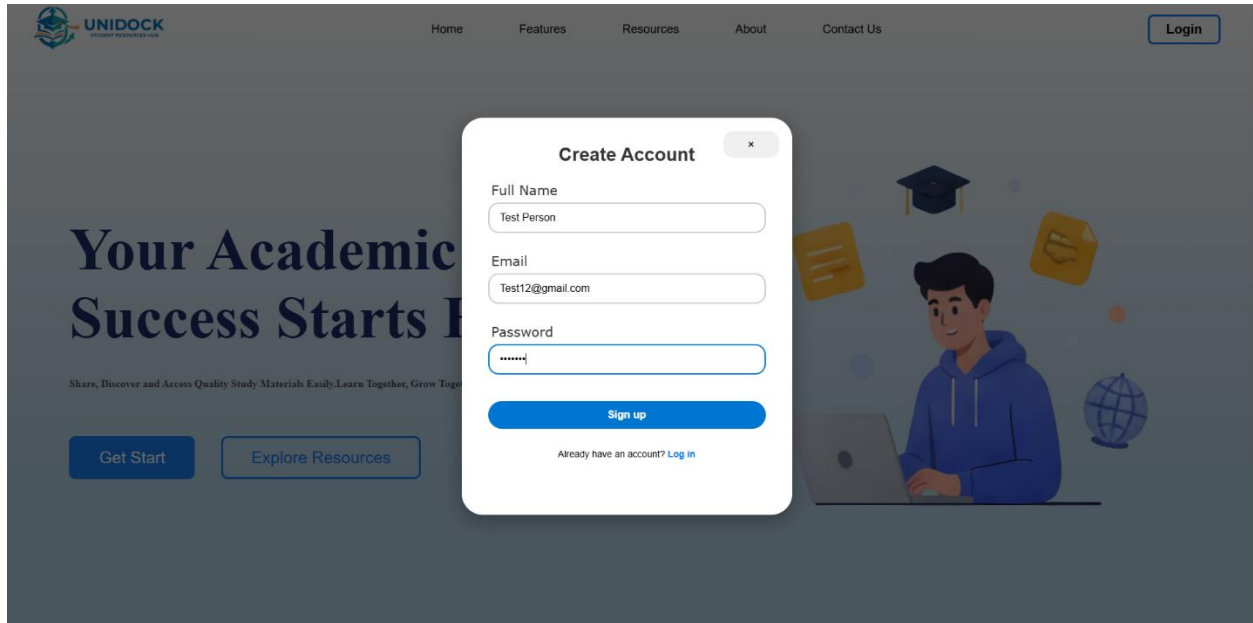
## 2.3 Folder Structure

```
Unidock/  
|-- auth/  
|   |-- login.php  
|   |-- logout.php  
|   |-- register.php  
|-- css/  
|   |-- about.css  
|   |-- Contact.css  
|   |-- features.css  
|   |-- home.css  
|   |-- Qa.css  
|   |-- resources.css  
|   |-- search.css  
|-- images/  
|-- includes/  
|   |-- db.php  
|   |-- functions.php  
|-- js/  
|   |-- features.js  
|   |-- login.js  
|   |-- Qa.js  
|   |-- resources.js  
|   |-- search.js  
|-- uploads/  
|-- about.php  
|-- bookmark.php  
|-- Contact.php  
|-- database.sql  
|-- features.php  
|-- index.php  
|-- Qa.php  
|-- README.md  
|-- reminder.php  
|-- resources.php  
|-- search.php  
|-- todo.php  
|-- upload.php
```

## 3. Implementation

### 3.1 User Authentication

Every protected page calls require login / is logged in from includes/functions.php. Registration stores a BCrypt hashed password with password hash, and the login modal posts to auth/login.php.



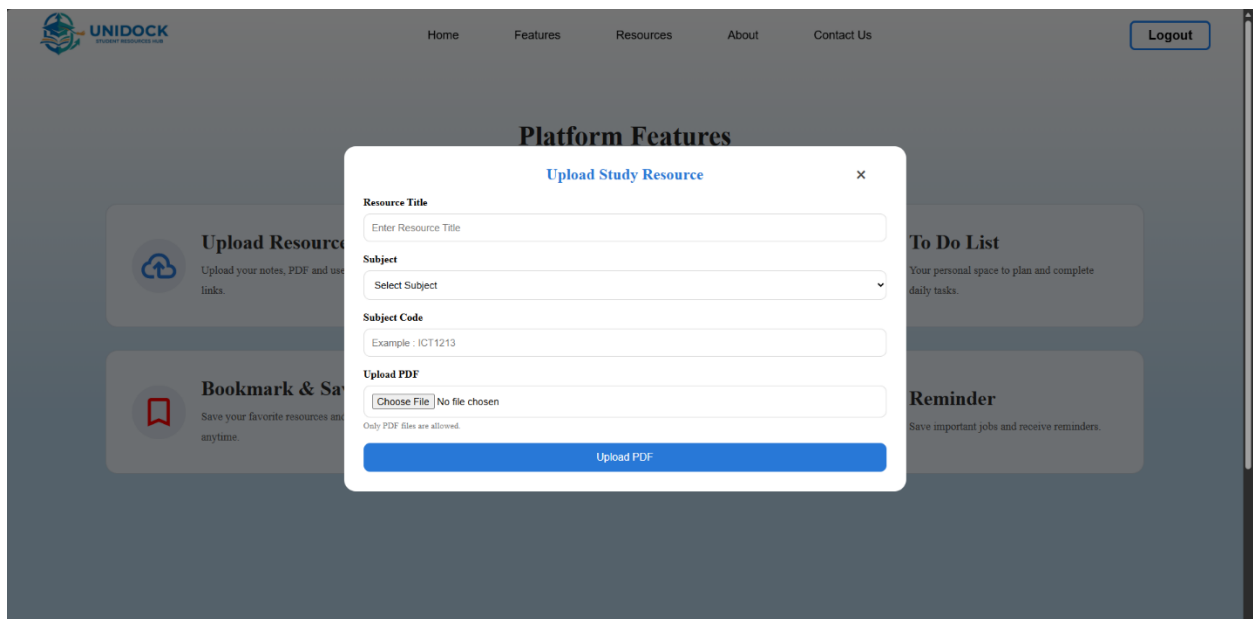
The screenshot shows the UNIDOCK website with a 'Create Account' modal open. The modal contains the following fields and buttons:

- Create Account** (Title)
- Full Name**: Text input field with 'Test Person' entered.
- Email**: Text input field with 'Test12@gmail.com' entered.
- Password**: Password input field with masked characters.
- Sign up**: Blue button.
- Already have an account? Log in**: Link.

The background shows the website header with 'Home', 'Features', 'Resources', 'About', 'Contact Us', and 'Login' buttons. The main content area has the text 'Your Academic Success Starts Here' and buttons for 'Get Start' and 'Explore Resources'.

### 3.2 Resource Upload

Upload.php validates that the logged in user supplied a title, subject and a file, confirms the files MIME type is application/pdf using , enforces a 10MB size limit, generates a collision-safe file name, and stores the file under /uploads/ before inserting a row into the resources table.



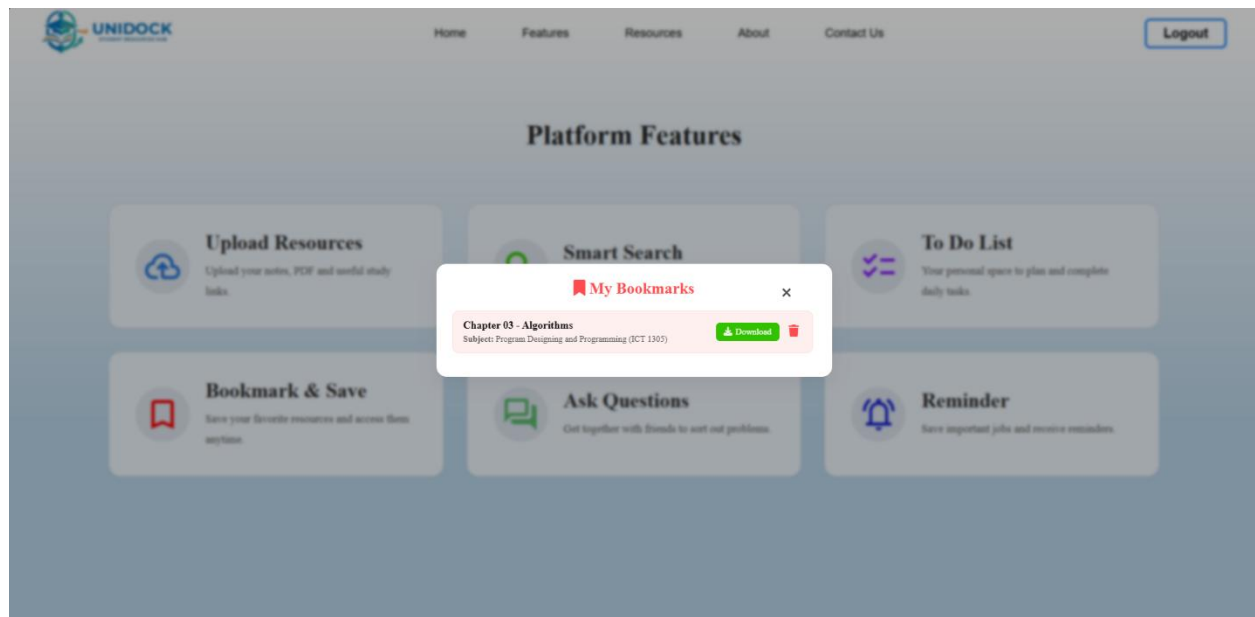
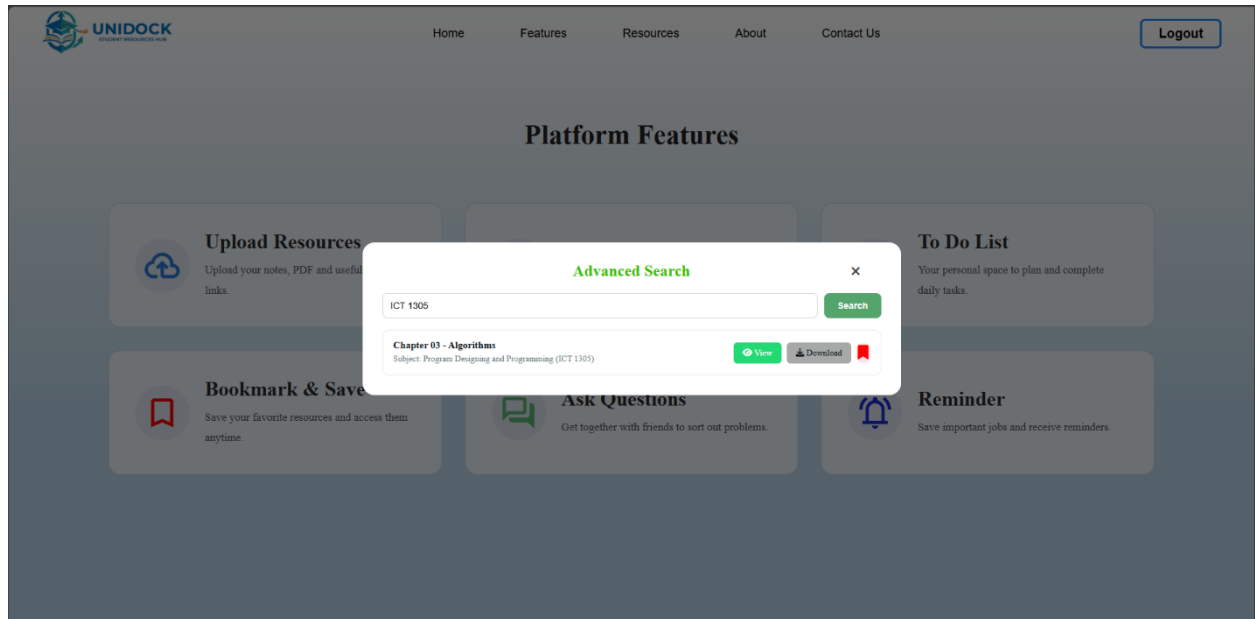
The screenshot shows the UNIDOCK website with a 'Platform Features' section and an 'Upload Study Resource' modal open. The modal contains the following fields and buttons:

- Upload Study Resource** (Title)
- Resource Title**: Text input field with 'Enter Resource Title' placeholder.
- Subject**: Dropdown menu with 'Select Subject' selected.
- Subject Code**: Text input field with 'Example : ICT1213' placeholder.
- Upload PDF**: Text input field with 'Choose File' button and 'No file chosen' text.
- Only PDF files are allowed.** (Text below the upload field)
- Upload PDF**: Blue button.

The background shows the website header with 'Home', 'Features', 'Resources', 'About', 'Contact Us', and 'Logout' buttons. The main content area has the text 'Platform Features' and sections for 'Upload Resource', 'Bookmark & Save', 'To Do List', and 'Reminder'.

### 3.3 Smart Search & Bookmarks

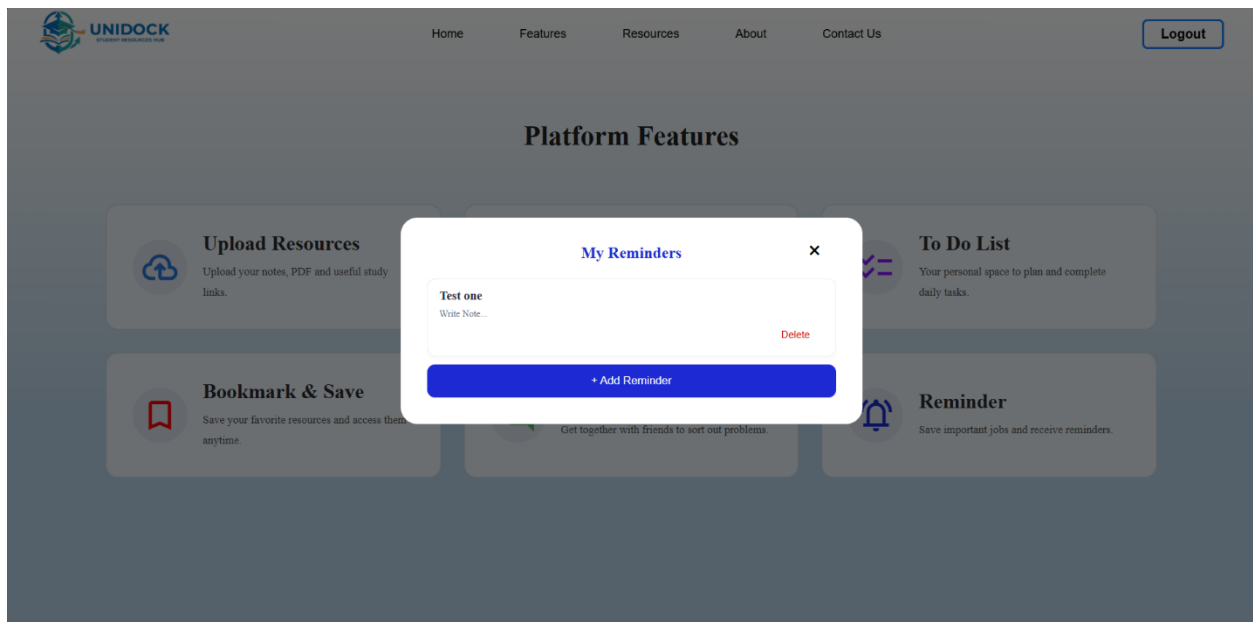
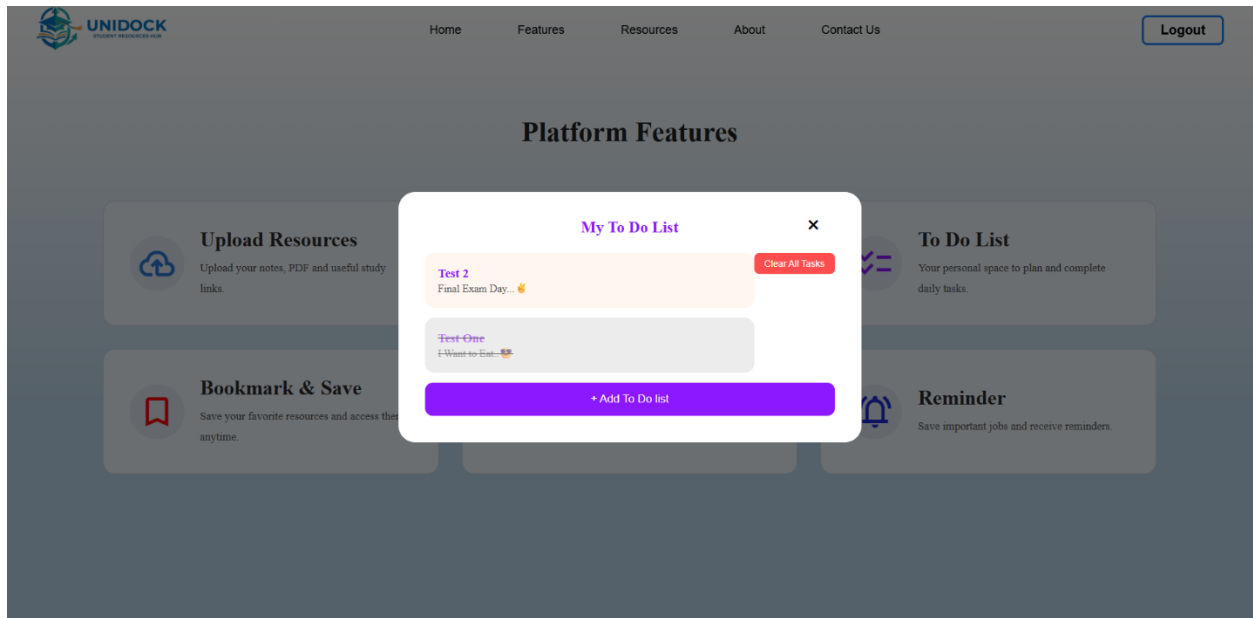
Search.php builds a parameterized SQL query that matches the keyword against title, subject, subject code and description, optionally filtered by subject, and returns the results as JSON together with each resources bookmark status for the current user. bookmark.php lets a logged in student toggle a bookmark or list all of their saved resources.





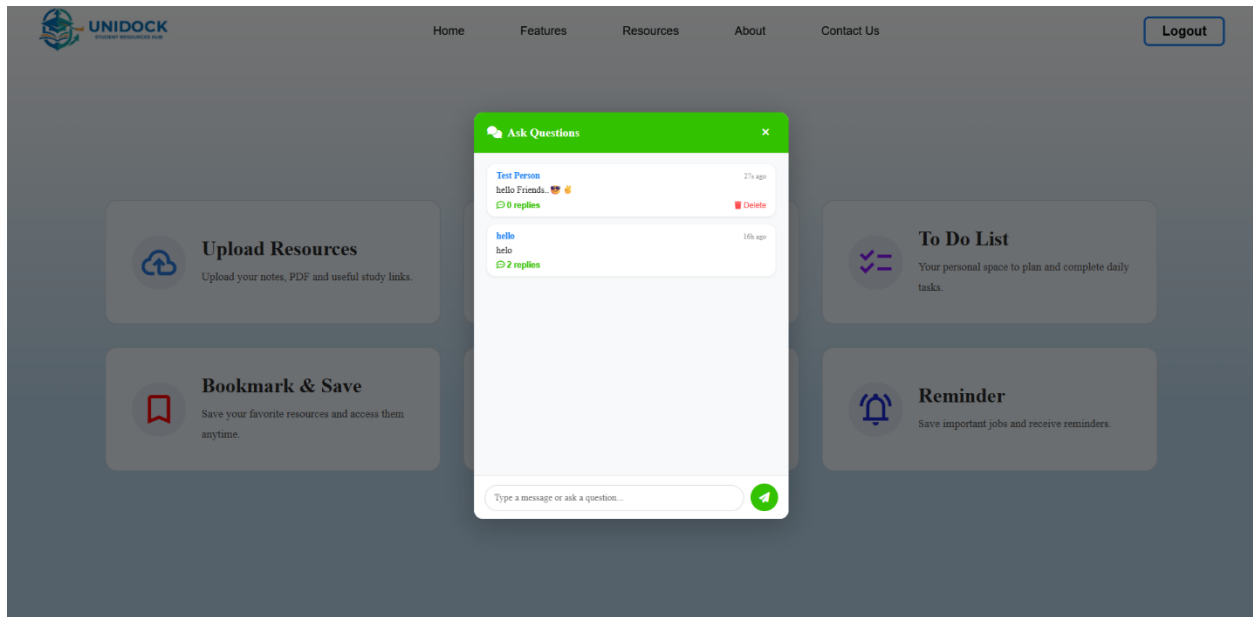
### 3.4 To Do List & Reminders

Todo.php and reminder.php are AJAX endpoints scoped to the logged in user that support listing, adding, toggling completion, deleting, and clearing tasks or reminders, each backed by its own MySQL table.



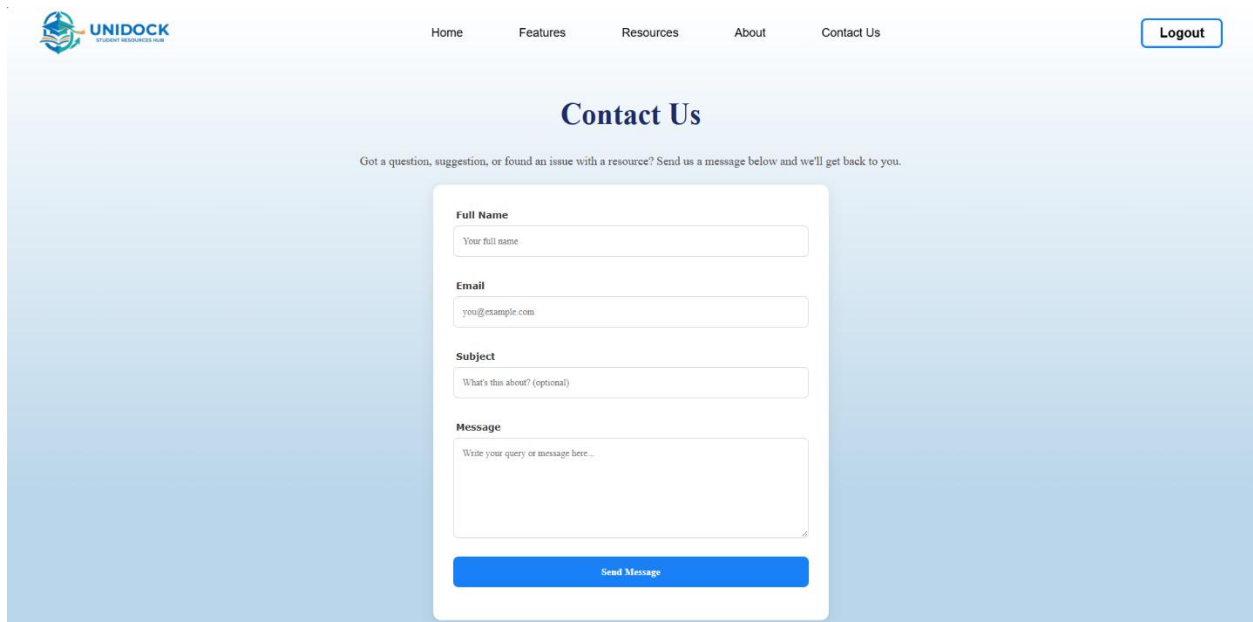
### 3.5 Ask a Question (Q&A)

Qa.php provides list, ask, view, reply, delete question and delete reply actions, joining the questions and question replies tables with the users table so each question and reply shows the authors name and reply count.



### 3.6 Contact Form

Contact.php validates the name, email and message fields, checks the email format with filter var(), and stores the submission in the messages table , before redirecting back with a success or error message.



## 4. Challenges and Solutions

| Challenge                                                      | Solution                                                                                                                                          |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Preventing SQL injection across many AJAX endpoints            | Used PDO prepared statements with bound parameters in every query, instead of concatenating user input into SQL.                                  |
| Restricting uploads to genuine PDF files                       | Checked the real MIME type with PHP extension and capped the upload size at 10MB in upload.php.                                                   |
| Keeping the to do list, reminders and Q&A private to each user | Every query in todo.php, reminder.php and Qa.php filters by the logged in users id from the session, so users only see and modify their own data. |
| Sharing one login/sign-up modal across many pages              | Built a single reusable modal with a hidden “redirect” field so it returns the user to the page they logged in from.                              |
| Making the layout work on mobile, tablet and desktop           | Used Bootstrap 5 grid and components together with custom responsive CSS and a collapsible mobile menu.                                           |

## 5. Individual Contribution

| Member                                                             | Responsibilities                                                                                                                                     |
|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Yasindu Sithum Nimsara<br>Reg No:- ITT/2024/074<br>Index No:- 2759 | UI/UX Design, Home page Frontend and Backend, Features page Frontend and Backend, Login Page Frontend and Backend.                                   |
| Dineth Induwara<br>Reg No:- ITT/2024/010<br>Index No:- 2695        | Resources Page Frontend and Backend, About Page Frontend and Backend, Contact Us Frontend and Backend and Bookmark & Save Feature, Reminder Feature. |

## 6. Setup Instructions

1. Install XAMPP and start the Apache and MySQL modules.
2. Copy the project folder into the htdocs directory (e.g. C:/xampp/htdocs/unidock).
3. Open phpMyAdmin, create a new database, then import database.sql or unidock\_exported\_sql.sql to create all tables.
4. Check includes/db.php and update the database host, name, username and password to match your local MySQL setup if needed.
5. Visit {http://localhost/unidock} in a browser to run the application.
6. Register a new account and log in to access the Features, Resources, To Do List, Reminders and Q&A sections.

## 7. References

- Bootstrap 5.3.3 Documentation – <https://getbootstrap.com>
- Font Awesome 6.7.2 – <https://fontawesome.com>
- I used it to further my learning of HTML, JS, and PHP. – <https://www.w3schools.com>